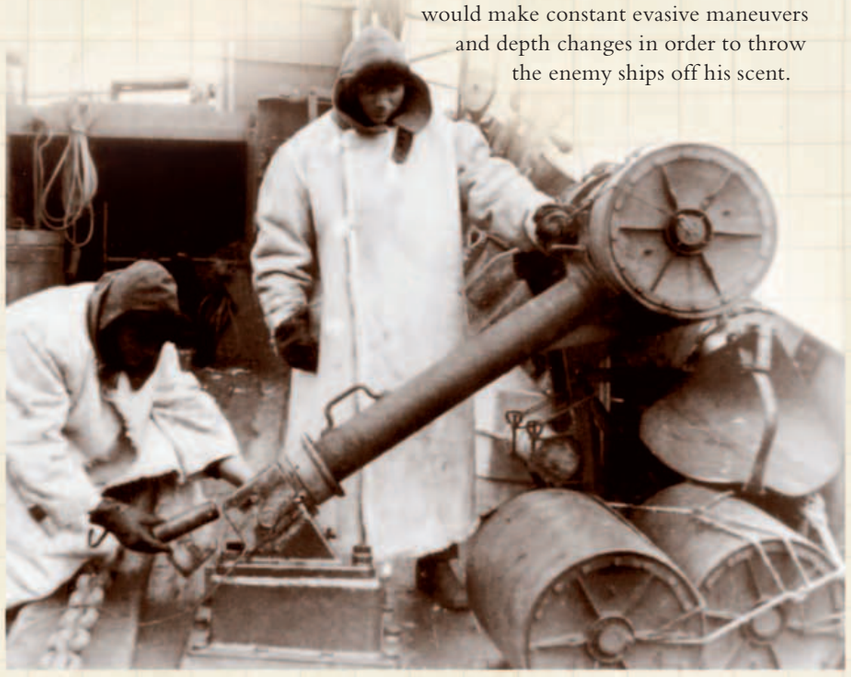
DESTROYING U-BOATS

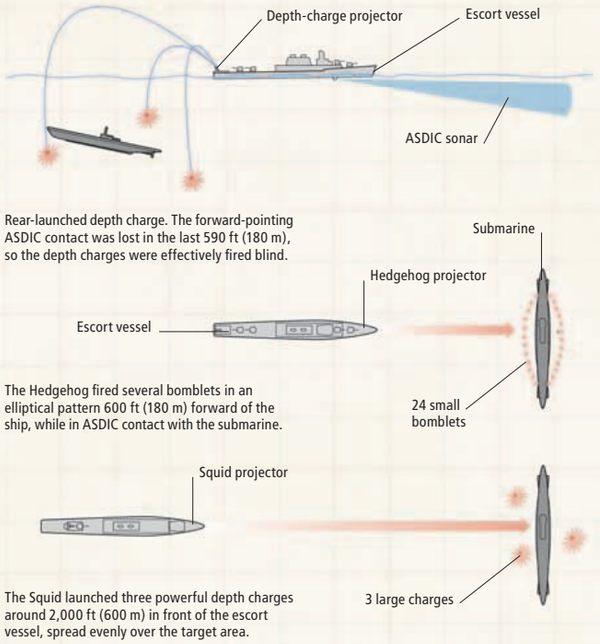
Once a convoy escort group had located a U-boat they had several means of engaging it. If the U-boat was on the surface, it was attacked with gunfire or torpedoes, or sometimes even rammed (maritime patrol aircraft would make strafing or depth-charge runs). In most cases, however, the U-boat would submerge. In this case, the primary weapon was the depth charge, a cylinder of explosive with a hydrostatic fuze that was set to detonate the weapon at a particular depth. Escort vessels would circle the known or predicted U-boat position and "bracket" the area with depth charges. Typically, an escort would drop three depth charges from the stern and fire one from each side in every attack, varying the depth settings of each depth charge to maximize the possibilities of a kill. Several ships working in tandem would be able to blanket a wide area with explosive. In response, the U-boat commander would make constant evasive maneuvers and depth changes in order to throw the enemy ships off his scent.



READY FOR LAUNCH A British naval crew prepare a depth charge for launch from a thrower. Depth charges may have been the crudest anti-submarine weapon in the Allied arsenal, but they were also extremely effective, accounting for 43 percent of all U-boats sunk.

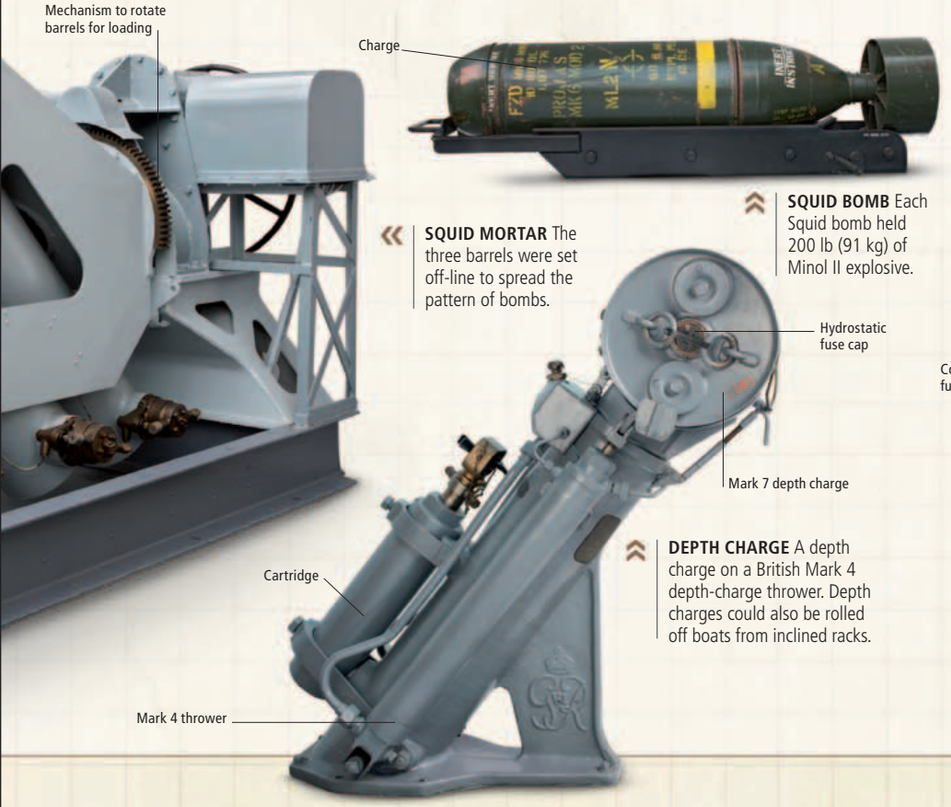
DEPTH-CHARGE LAUNCH PATTERNS

As the war progressed, two main forward-firing variations on the depth charge emerged. The Squid three-barreled mortar fired three fast-sinking depth charges that produced triangular explosive patterns beneath the water, increasing the chances of a successful kill. The Hedgehog, by contrast, fired 24 small contact-detonated bomblets over a wide circular area.



ANTI-SUBMARINE DEPTH CHARGES AND MORTARS

The limitations of British sonar (ASDIC) were critical in the development of new anti-submarine weaponry during World War II. ASDIC was forward pointing, and therefore lost contact with a U-boat if it was within 600 ft (180 m) of the front of the ship. New types of anti-submarine weapon such as the Hedgehog and Squid mortars, introduced in 1942–43, fired forward and far enough ahead to engage the U-boat while still within ASDIC contact. The shape of their bombs also meant that they sank faster than depth charges, so escort vessels could put their explosives on target more quickly if a contact was made.



SQUID MORTAR The three barrels were set off-line to spread the pattern of bombs.

SQUID BOMB Each Squid bomb held 200 lb (91 kg) of Minol II explosive.

DEPTH CHARGE A depth charge on a British Mark 4 depth-charge thrower. Depth charges could also be rolled off boats from inclined racks.



HEDGEHOG BOMBLET Four rows of six spigots launched a salvo of 24 spigot-mortar bomblets.